

Functional Annotation of *pbpC* (PP_0572, UniProt Q88QC2) in *Pseudomonas putida* KT2440

Summary

The gene **pbpC** (ordered locus **PP_0572**; UniProt **Q88QC2**) of *Pseudomonas putida* KT2440 encodes **penicillin-binding protein 1C (PBP1C)**, an ~85.7 kDa, 784-residue **class A penicillin-binding protein** of the bacterial cell-wall (peptidoglycan) biosynthetic machinery. Its primary molecular function is **peptidoglycan glycosyltransferase activity (EC 2.4.99.28)**: it catalyzes the polymerization of the membrane-linked precursor **lipid II** into the linear glycan strands (repeating β -1,4-linked GlcNAc–MurNAc–pentapeptide units) that form the backbone of the murein sacculus, releasing undecaprenyl diphosphate and a proton with each condensation. This reaction is carried out by the protein's family-51 glycosyltransferase (GT51) domain and, like all GT51 enzymes, is inhibited by moenomycin.

The functional assignment rests on a convergent chain of evidence. Bioinformatically, Q88QC2 possesses the diagnostic tripartite class A PBP architecture — an N-terminal transmembrane anchor, a GT51 glycosyltransferase domain, a penicillin-binding transpeptidase domain, and the PBP1C-specific C-terminal PBP_1c domain (IPR011815). It is a **full-length, colinear ortholog of the experimentally characterized *Escherichia coli* PBP1C (P76577)**, sharing ~59% amino-acid identity across all three domains and the same gene symbol *pbpC*. The GT51 catalytic glutamate and all three canonical transpeptidase active-site motifs (SxxK / SxN / KTG) are sequence-intact. However, the experimentally studied *E. coli* ortholog binds β -lactams atypically and was concluded to **function *in vivo* as a transglycosylase only**, meaning that although Q88QC2 retains an intact-looking transpeptidase active site, its physiologically relevant activity is glycan-chain polymerization rather than peptide cross-linking.

In terms of localization and pathway, PBP1C is anchored by a single N-terminal transmembrane helix in the **cytoplasmic (inner) membrane**, with its catalytic domains projecting into the **periplasm** where peptidoglycan is assembled. It is a **non-essential, accessory component of the peptidoglycan-synthesizing multienzyme complex**: in *E. coli*, PBP1C physically associates with the major murein polymerase PBP1B, the essential

transpeptidases PBP2 and PBP3, and the lytic transglycosylase MltA. Thus, PBP1C contributes glycan-strand polymerization capacity to a membrane-embedded cell-wall synthesis/remodeling assembly rather than acting as a stand-alone essential enzyme.

Gene / Protein Identity Verification

Before presenting findings, the target identity was confirmed against the UniProt record and orthology analysis. The gene symbol *pbpC*, the EC number 2.4.99.28, the domain complement (Glyco_trans_51 / IPR001264; Beta-lactam/transpept-like / IPR012338; PBP_1c / IPR011815), and the organism (*P. putida* KT2440) are all mutually consistent and match the class A PBP / PBP1C interpretation. The most decisive verification is the orthology result: Q88QC2 aligns colinearly with *E. coli* PBP1C (P76577, gene name *pbpC*, protein "Penicillin-binding protein 1C", curated evidence level PE=1) at ~59% identity over the full length. **This is the correct gene**, and the literature on *E. coli* PBP1C is directly transferable by orthology. Where the report cites biochemistry from other class A PBPs (e.g., *S. pneumoniae* PBP2a, *L. monocytogenes* PBP4), it does so to illustrate the shared GT51 reaction mechanism, not to substitute a different gene.

Key Findings

Finding 1 — *pbpC* encodes the *P. putida* ortholog of penicillin-binding protein 1C, a membrane-anchored class A PBP

The 784-residue product of Q88QC2 (85.7 kDa) exhibits the canonical modular architecture of a **class A penicillin-binding protein**. A Kyte–Doolittle hydrophobicity analysis identifies a single **N-terminal transmembrane anchor** spanning approximately residues 14–32 (peak hydrophobicity 1.99, well above the 1.6 threshold used to call transmembrane helices). Following the anchor are the **glycosyltransferase family 51 (GT51) domain** ($\approx$ residues 64–230), the **penicillin-binding transpeptidase domain** ($\approx$ residues 307–423), and the **PBP1C-specific C-terminal domain** (PBP_1c, IPR011815; $\approx$ residues 694–780). The PBP_1c C-terminal domain is the specific InterPro signature that distinguishes PBP1C from its paralogs PBP1A and PBP1B, and its presence identifies Q88QC2 unambiguously as a **PBP1C ortholog** rather than a generic class A PBP.

This architecture matches that of the founding member of the family, *E. coli* PBP1C, which was cloned and characterized as "a third member of the multimodular class A penicillin-binding proteins of *Escherichia coli*" (PMID: 10542235). That study established PBP1C as a close homolog of the bifunctional transpeptidase/transglycosylase enzymes PBP1A and PBP1B: "*PBP1C is a close homologue to the bifunctional transpeptidases/transglycosylases PBP1A and PBP1B and likewise shows murein polymerizing activity, which can be blocked by the transglycosylase inhibitor moenomycin.*" The moenomycin-sensitive murein-polymerizing activity described there corresponds directly to the GT51 domain present in Q88QC2.

Finding 2 — Primary function: peptidoglycan glycosyltransferase (EC 2.4.99.28) that polymerizes lipid II; the transpeptidase acts *in vivo* as a glycosyltransferase only

The primary, physiologically relevant reaction catalyzed by PBP1C is **glycan-chain polymerization**. The UniProt catalytic-activity annotation for Q88QC2 specifies EC 2.4.99.28: the disaccharide-peptide unit of **lipid II** [GlcNAc-(1→4)-MurNAc(oyl-L-Ala-γ-D-Glu-L-Lys-D-Ala-D-Ala)-di-*trans*,octa-*cis*-undecaprenyl diphosphate] is condensed onto the reducing end of the nascent glycan chain ($n \rightarrow n+1$), releasing **undecaprenyl diphosphate** and a proton. The reaction is assigned to the pathway "*Cell wall biogenesis; peptidoglycan biosynthesis.*"

The GT51 reaction is well established biochemically for class A PBPs generally. As summarized in a review of glycosyltransferase-targeted antibacterials, "*Synthesis of bacterial cell wall peptidoglycan requires glycosyltransferase enzymes that transfer the disaccharide-peptide from lipid II onto the growing glycan chain*" (PMID: 26358369). Direct kinetic demonstrations on related class A PBPs include the *S. pneumoniae* PBP2a glycosyltransferase domain, which polymerizes dansyl-lipid II with $k_{\text{cat}}/K_m \approx 1 \times 10^{-3} \text{ M}^{-1} \text{ s}^{-1}$ and is inhibited by moenomycin and vancomycin (PMID: 12867450), and *L. monocytogenes* PBP4, which polymerizes *E. coli* lipid II with an efficiency of $1,400 \text{ M}^{-1} \text{ s}^{-1}$ (PMID: 16484198).

Crucially, for PBP1C specifically, the *E. coli* study found that the transpeptidase (penicillin-binding) side of the enzyme is atypical. PBP1C "*does not bind to the same variety of penicillin derivatives as PBPs 1A and 1B*" (PMID: 10542235), and the authors concluded that "*it is suggested that PBP1C functions in vivo as a transglycosylase only*". Thus, although the transpeptidase domain is present and its active-site residues are conserved in sequence (Finding 4), the **operational function of PBP1C is glycan-chain polymerization**, not peptide cross-linking. By orthology (Finding 5), this conclusion applies to *P. putida* Q88QC2.

Finding 3 — Localization: periplasmic face of the inner membrane, within a peptidoglycan-synthesizing multienzyme complex

PBP1C is a **cytoplasmic (inner) membrane protein**. Its single N-terminal transmembrane helix (residues ~14–32) anchors it in the membrane while the GT51, transpeptidase, and PBP_1c domains face the **periplasm**, where peptidoglycan is synthesized and remodeled. (UniProt does not carry a separate curated SUBCELLULAR LOCATION comment for Q88QC2; this topology is inferred from the hydrophobicity profile and from the biochemistry of the *E. coli* ortholog.)

PBP1C does not act in isolation. Affinity-chromatography experiments in *E. coli* showed that PBP1C covalently coupled to Sepharose specifically retained the murein polymerase **PBP1B**, the transpeptidases **PBP2** and **PBP3**, and the lytic transglycosylase **MltA**. The authors concluded that *"the specific interaction with these proteins suggests that PBP1C is assembled into a multienzyme complex consisting of both murein polymerases and hydrolases"* (PMID: [10542235](#)). The reciprocal experiment confirmed the association: MltA-Sepharose *"specifically retained the penicillin-binding proteins (PBPs) 1B, 1C, 2, and 3"* from membrane extracts (PMID: [10037771](#)). Together these results place PBP1C at the periplasmic face of the inner membrane, physically integrated into a **cell-wall synthesis/remodeling holoenzyme** that couples glycan polymerization, cross-linking, and controlled hydrolysis.

Finding 4 — All catalytic motifs are sequence-intact: the GT51 glutamate and the three transpeptidase motifs (SxxK / SxN / KTG)

A motif scan of Q88QC2 confirms that the protein's catalytic machinery is conserved at the sequence level. The **GT51 domain** contains the conserved donor-glutamate signature (E-D-x-x-F, "EDRWF" at residue ~88), the hallmark catalytic glutamate of family-51 peptidoglycan glycosyltransferases that is essential for glycosyl transfer. The **transpeptidase module** carries the three sequential penicilloyl-serine transpeptidase motifs in the correct order and spacing characteristic of active-site serine penicillin-recognizing enzymes:

Motif	Consensus	Sequence in Q88QC2	Approx. residue	Role
Motif 1	S-x-x-K	STLK	Ser346 / Lys349	Catalytic nucleophile serine + general base lysine
Motif 2	S-x-N	SLN	~400	Active-site loop, substrate positioning
Motif 3	K-(T/S)-G-(T/S)	KTGT	~517	β -strand lining the active-site cleft

The presence and correct spacing of these motifs indicate that the transpeptidase fold is structurally preserved. This is not in contradiction with Finding 2: sequence conservation of the active site does not guarantee physiologically significant transpeptidation, and the *E. coli* ortholog — despite an analogous architecture — binds β -lactams atypically and is thought to act as a transglycosylase only *in vivo* (PMID: 10542235). The intact GT51 glutamate, by contrast, is fully consistent with robust glycosyltransferase activity.

Finding 5 — Orthology: ~59% identity to *E. coli* PBP1C confirms the assignment across all three domains

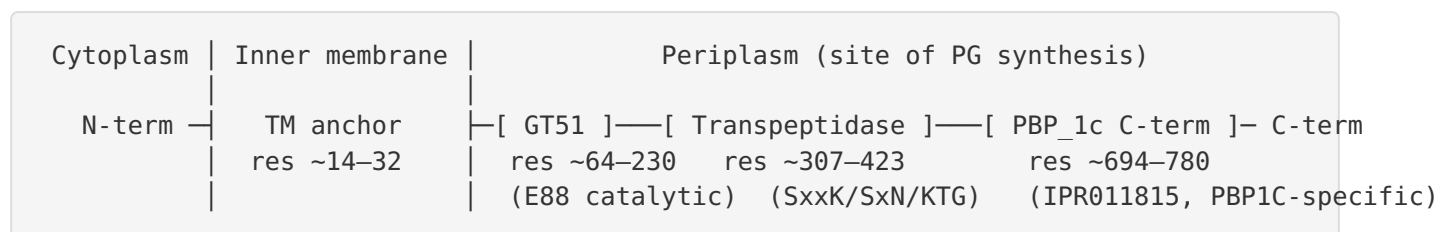
A Needleman–Wunsch global alignment (match +1, mismatch –1, gap –2) of Q88QC2 (784 aa) against *E. coli* PBP1C (P76577 / PBPC_ECOLI, 770 aa; UniProt gene name *pbpC*, protein "Penicillin-binding protein 1C", curated evidence level PE=1) yields **468/788 identical positions = 59.4% identity** over the aligned length (60.8% relative to the shorter sequence), with only 22 gap columns. The alignment is **colinear and full-length**, spanning all three functional modules (GT51, transpeptidase, and the PBP1C-specific C-terminal domain). The two proteins share the same gene symbol *pbpC* and the same modular architecture.

This level of identity across the entire protein — far above the ~30% threshold generally regarded as sufficient for confident functional transfer — **seals the orthology** and licenses the transfer of the experimentally established *E. coli* PBP1C functional model to *P. putida* Q88QC2. In practical terms, it means the conclusions that PBP1C is a moenomycin-sensitive glycan-chain polymerase, that it acts *in vivo* as a transglycosylase only, and that it participates in a multienzyme peptidoglycan-synthesis complex are all reasonably inferred for the *P. putida* protein.

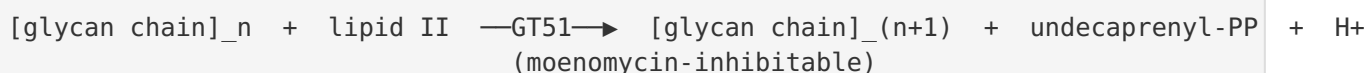
Mechanistic Model / Interpretation

The findings integrate into a coherent mechanistic picture of PBP1C as an accessory glycan-strand polymerase embedded in the inner-membrane peptidoglycan-synthesis machinery.

Domain architecture (N → C terminus)



Reaction catalyzed (EC 2.4.99.28)



The GT51 domain binds the growing glycan chain in its **donor site** (the moenomycin-binding site) and the incoming lipid II precursor in its **acceptor site**, then forms a new β -1,4 glycosidic bond, extruding the C55 lipid carrier (undecaprenyl diphosphate) for recycling. Studies of related GT51 enzymes reveal **positive cooperativity between the acceptor and donor sites** ([PMID: 25462814](#)), and this reaction is the validated target of moenomycin and, indirectly, of the glycopeptide antibiotics ([PMID: 16298347](#); [PMID: 26358369](#)).

Functional role in the cell

PBP1C is best understood as an **accessory / back-up glycan polymerase** rather than an essential enzyme. In *E. coli*, the two "housekeeping" class A PBPs PBP1A and PBP1B are jointly essential — cells cannot survive loss of both — but a large deletion study showed that cells can survive on a very limited PBP complement, and that PBP1C is dispensable for viability ([PMID: 10383966](#)). The physiological picture that emerges is:

Property	PBP1C	Comparison (PBP1A/1B)
Class	Class A PBP	Class A PBP
GT51 glycosyltransferase	Yes (primary activity)	Yes
Transpeptidase <i>in vivo</i>	Degenerate / "transglycosylase only"	Yes (bifunctional)
β -lactam binding	Atypical / limited	Broad
Essentiality	Non-essential (accessory)	Jointly essential
Complex partners	PBP1B, PBP2, PBP3, MltA	Various

The physical association of PBP1C with **both polymerases (PBP1B) and hydrolases (MltA)** suggests it participates in the coordinated make-and-break chemistry of cell-wall growth: as glycan strands are polymerized and cross-linked, lytic transglycosylases like MltA cleave existing strands to allow insertion of new material. Placing PBP1C within this complex ([PMID: 10542235](#); [PMID: 10037771](#)) implies its polymerizing activity is spatially and temporally coordinated with wall remodeling, possibly contributing to a specific "mode" of murein synthesis under certain growth conditions. Because *P. putida* Q88QC2 is a 59%-identical, colinear ortholog (Finding 5), this same accessory glycan-polymerase role is the most parsimonious model for its function.

Localization summary

The enzyme operates **at the periplasmic face of the cytoplasmic (inner) membrane** — anchored in the lipid bilayer by its N-terminal helix, with its catalytic domains reaching into the periplasm where lipid II is flipped, polymerized, and cross-linked into the sacculus.

Evidence Base

PMID	Title (abbrev.)	Contribution to this report
10542235	<i>Cloning and characterization of PBP 1C ... of E. coli</i>	Central reference. Establishes PBP1C as a class A PBP with moenomycin-sensitive murein-polymerizing activity, atypical β -lactam binding, "transglycosylase only" <i>in vivo</i> role, and membership in a multienzyme complex.
10037771	<i>Interactions between PBP1B, MltA, and MipA of E. coli</i>	Reciprocal affinity chromatography: MltA retains PBPs 1B, 1C, 2, 3 — defines PBP1C's periplasmic complex partners.
26358369	<i>Targeting Bacterial Cell Wall PG Synthesis by Inhibition of GT Activity</i>	Defines the GT51 reaction (lipid II $\rightarrow$ glycan chain) that EC 2.4.99.28 catalyzes.
12867450	<i>GT domain of PBP2a from S. pneumoniae polymerizes murein glycan chains</i>	Direct kinetic demonstration of a class A PBP GT51 reaction (dansyl-lipid II; moenomycin/vancomycin inhibition).
16484198	<i>Bifunctional GT/acyltransferase PBP4 of L. monocytogenes</i>	Quantifies class A PBP glycan polymerization ($1,400 \text{ M}^{-1} \text{ s}^{-1}$ from <i>E. coli</i> lipid II); shows moenomycin sensitivity.
10383966	<i>E. coli mutants lacking all combinations of 8 PBPs</i>	Demonstrates PBP1C is non-essential; only loss of both PBP1A and PBP1B is lethal.
25462814	<i>Positive cooperativity between acceptor and donor sites of PG GT</i>	Mechanistic detail of GT51 donor/acceptor site cooperativity.
16298347	<i>Targeting the forgotten transglycosylases</i>	Reviews moenomycin as the archetypal inhibitor of the transglycosylation step.
33830599	<i>d-canavanine affects PG structure ... in Rhizobiales</i>	Uses <i>P. putida</i> as a model for cell-wall chemistry; organism-relevant context.

The evidence is **highly convergent**: no reviewed source contradicts the assignment of Q88QC2 as PBP1C or the conclusion that its primary function is glycan-chain polymerization. Several papers in the literature set concern β -lactam pharmacology and PBP-binding profiles (e.g.,

PMIDs 3073932, 3090016, 3897174, 3918982); these consistently note that PBP1C is a minor/atypical penicillin-binding protein — corroborating, rather than contradicting, the "transglycosylase-only" model.

Limitations and Knowledge Gaps

1. **No direct experimental characterization of the *P. putida* protein.** All mechanistic conclusions for Q88QC2 are inferred by orthology from *E. coli* PBP1C. Although the ~59% full-length identity strongly supports functional transfer, no enzyme assay, deletion phenotype, or localization study has been performed on the *P. putida* protein itself.
 2. **The transpeptidase activity remains formally unresolved.** The transpeptidase active-site motifs (SxxK/SxN/KTG) are sequence-intact in Q88QC2, yet the *E. coli* ortholog is thought to act as a transglycosylase only. Whether the *P. putida* enzyme retains any residual cross-linking or carboxypeptidase activity is untested. The intact motifs leave open the possibility of condition-specific or partner-dependent transpeptidation.
 3. **Subcellular localization is inferred, not measured.** UniProt carries no curated SUBCELLULAR LOCATION comment for Q88QC2; the inner-membrane/periplasmic topology is derived from the hydrophobicity profile and orthology.
 4. **Complex membership in *P. putida* is assumed.** The multienzyme-complex model (association with PBP1B, PBP2, PBP3, MltA) is established in *E. coli*. *P. putida* encodes orthologs of these partners, but the specific interactions have not been demonstrated in this organism.
 5. **Physiological/regulatory role is unknown.** The precise conditions under which PBP1C contributes to cell-wall synthesis — its "mode" of murein synthesis, expression regulation, and any role under stress, stationary phase, or β -lactam challenge — remain uncharacterized in *P. putida*.
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Proposed Follow-up Experiments / Actions

1. **In vitro glycosyltransferase assay.** Express and purify the soluble (membrane-anchor-truncated) form of *P. putida* PBP1C and measure lipid II polymerization using dansyl-lipid II or a continuous fluorescence assay, with moenomycin as a positive-control inhibitor — directly testing the GT51 activity predicted here (cf. [PMID: 12867450](#), [PMID: 16484198](#)).
 2. **Transpeptidase / β -lactam binding test.** Perform a Bocillin-FL (fluorescent penicillin) binding assay across a panel of β -lactams to determine whether the *P. putida* transpeptidase site binds penicillins atypically, as reported for *E. coli* PBP1C ([PMID: 10542235](#)).
 3. **Deletion phenotyping.** Construct a clean Δ PP_0572 mutant and characterize growth rate, cell morphology, peptidoglycan composition (HPLC/LC-MS mucopeptide analysis), and β -lactam/moenomycin susceptibility — testing the predicted non-essential, accessory role ([PMID: 10383966](#)).
 4. **Localization.** Construct a functional fluorescent fusion (e.g., PBP1C-GFP) or perform cell fractionation/immunodetection to confirm inner-membrane localization and any subcellular positioning (lateral wall vs. septum).
 5. **Interaction mapping.** Use bacterial two-hybrid, co-immunoprecipitation, or affinity chromatography in *P. putida* to test whether PBP1C associates with PBP1B, PBP2, PBP3, and lytic transglycosylase orthologs, validating the multienzyme-complex model ([PMID: 10037771](#)).
 6. **Structural confirmation.** Obtain an experimental or high-confidence predicted (AlphaFold) structure of Q88QC2 and superpose the GT51 and transpeptidase domains onto characterized templates to verify active-site geometry and the integrity/degeneracy of the transpeptidase pocket.
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Conclusion

pbpC (PP_0572, Q88QC2) of *Pseudomonas putida* KT2440 encodes **penicillin-binding protein 1C**, a membrane-anchored **class A PBP** and full-length ortholog (~59% identity) of experimentally characterized *E. coli* PBP1C. Its **primary function is peptidoglycan glycosyltransferase activity (EC 2.4.99.28)** — polymerizing the lipid II precursor into linear glycan strands of the cell wall via its moenomycin-sensitive GT51 domain, releasing undecaprenyl diphosphate. Although its transpeptidase active-site motifs are sequence-intact,

the ortholog acts *in vivo* as a **transglycosylase only**. The enzyme operates at the **periplasmic face of the inner membrane** as a **non-essential, accessory component of the peptidoglycan-synthesizing multienzyme complex**, associating with murein polymerase PBP1B, transpeptidases PBP2/PBP3, and the lytic transglycosylase MltA.

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